Loss of ergativity in Eastern Indo-Aryan languages

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Old Eastern Indo-Aryan stage/OEIA (900–1100 AD approx) showed split ergativity with evidence from its sole source *Caryapada*. New Eastern Indo-Aryan/EIA languages have lost their ergative properties (Stump, 1983; Bubenik, 1989, among others) gradually, evolving as nominative-accusative languages in the present time. The current study highlights how the Voice head in the structure is responsible for the evolution of the case system in the EIA languages. As the Voice head is intricately related to the external theta role and ergative case (Legate, 2014), it is essential to note its characteristics in the diachrony of EIA languages. This paper proposes that OEIA ergativity, seen in *Caryapada*, is DP-ergative, following the diagnostics of Polinsky (2016), where the Voice head assigns structural ergative to subject DP. It entails that OEIA has a distinct Voice head in its spine, which assigns the structural ergative case to the DP. The diagnostics used are agreement, A-movement, A′ movement, Binding, and coordinate structures. However, such split ergativity is not seen in the present EIA languages. This entails that the Voice head is lost. This loss led to the rise of a TP-determined structure for all tenses, leading to the emergence of the nominative alignment.

Keywords: ergativity, ergativity loss, voice, Eastern Indo-Aryan Languages

1. Introduction

This paper discusses the diachronic presence and subsequent loss of a Voice[1] head and produces evidence from the structure of early Eastern Indo-Aryan Languages (hereafter, EIAL) in *Caryapada* literature. EIALs are the eastern off-shoot of the Indo-Aryan Language family (hereafter, IAL), which are primarily spoken

1. Voice, in our work, is a functional head and this is represented with an upper case V in Voice following Legate (2014).

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in the eastern part of India and Bangladesh. *Caryapada* is the common source text of all present EIALs. *Caryapada* is an eighth – twelfth century text, which is a collection of poems by twenty-three poets of that time. The EIALs used in the study are Assamese, Bangla, and Odia, which are the official languages of the Indian states of Assam, West Bengal, and Odisha, respectively. According to Singh (2009), Mayadhar (1962), and Chatterji (1926), the *Caryapada* text is part of the Early Assamese, Old Odia, and Old Bangla periods themselves. *Caryapada* shows ergative case patterns in contrast to the present-day EIALs. This paper discusses how the EIALs have lost this ergativity to become nominative-accusative patterns.

Loss of ergativity in EIALs has been discussed in the literature, such as Stump (1983), who lists eight reasons that assisted in ergativity erosion in EIALs. These included "total suppletion of the ergative by the nominative", suppletion of the original nominative, and addition of the pronominal clitics or endings to verbal agreement, etc. Bubenik (1989) further adds three more reasons for the weakening or loss of ergativity, viz., the appearance of the absolutive case at the end of the MIA period, the divergent development of the ergative and the passive construction, and the consequences of the cliticization of the pronominal suffixes and the copula to the source of ergative in IALs – the *-ta* form. Stronski (2009, p. 105) also points out that the 'substratum effect' of Munda and Dravidian languages is responsible for the loss of ergativity. However, none of the works talks about the clause structure related to ergativity, as seen in *Caryapada*, and its loss, as seen in Late Middle Bangla (hereafter, LMB), Old Odia, and Early Assamese.

This paper investigates two questions. First, we probe into the properties of the ergativity in *Caryapada* (Early EIAL). Further, it explains how the ergativity was lost. This paper claims that diachronically, the Voice head was responsible for the licensing of nominals by the structural ergative case in the non-present tenses. The empirical evidence focuses on the diachrony of Bangla because Old Assamese and Old Odia have lost the original Voice head, seen in earlier sources of *Caryapada*, without documentation of the loss. Bangla diachrony, however, documents the change. The paper is arranged as follows: Section 1 introduces the problem. Section 2 briefly discusses the diachronic stages of 3 EIALs- Assamese, Odia, and Bangla. Section 3 elaborates on *Caryapada*, the lone text of the early stage of EIAL. This stage is particularly important because it produces evidence of the Voice head. This section also shows empirical evidence of how ergativity was subsequently lost. Section 4 connects the data with the presence and absence of Voice Phrase and Voice head. We conclude our study in Section 5.

2. Diachronic stages of Eastern Indo-Aryan languages

2.1 Diachronic stages

This section briefly introduces all the diachronic stages of the three languages in hand- Assamese, Odia, and Bangla in Table (1–3). Table 1 shows the diachronic stages of the Assamese language, starting from Early Assamese to Modern Assamese. Similarly, Table 2 records the diachronic stages of the Odia language from the Old Odia stage to Modern Odia. Table 3 shows the diachronic stages of Bangla from Old Bangla to Modern Bangla.

Table 1. Diachronic stages of EIA language assamese

Stages	Dates (Singh, 2009)	Dates (Devi, 1986, Kakati, 1941)
Early Assamese Period	6–1500 AD	14–1600 AD
Middle Assamese Period	17–1900 AD	17–1900 AD
Modern Assamese	Late 1900 AD	1900 AD

Table 2. Diachronic stages of EIA language Odia

Stages	Dates (Mayadhar, 1962)	Dates (Mishra, 1997)
Old Odia	8–1300 AD	11–1500 AD
Early Middle Odia	13–1500 AD	16–1850 AD
Middle Odia	15–1700 AD	
Late Middle Odia	17–1850 AD	
Modern Odia		From 1850 AD

Table 3. Diachronic stages of EIA language Bangla

Stages	Dates (Chatterji, 1926)
Formative or Old Bangla	900–1200 AD
Transitional Age / Dark Age	1200–1300 AD
Early Middle Bangla	1300–1500 AD
Late Middle Bangla	1500–1800 AD
Modern Bangla	1800- present

After discussing the evolution of the EIALs, a rudimentary question about the source of the EIALs arises. The next section discusses the earliest literary source of the EIALs.

3. Source of eastern Indo-Aryan languages: *Caryapada*

This section discusses the typological characteristics of the EIALs and discusses their oldest literary source, *Caryapada*.

3.1 Eastern Indo-Aryan languages

EIALs descend from Magadhan Prakrit, one of the literary Prakrits in the 2nd Middle Indo-Aryan (MIA) period (200–600AC). In the Late MIA period (600–1000 AD), the languages were largely divided into two major dialects: the literary Apabhramsa (later part of Sauraseni Prakrit) and the Magadhan languages (see Chatterji, 1926 for a discussion). *Caryapada* is taken to be the source of all Magadhan languages or present EIALs, which are divided into three groups as listed below.

i. Eastern Magadhan: Bangla, Assamese, Odia
ii. Central Magadhan: Maithili, Magahi
iii. Western Magadhan: Bhojpuriya

EIALs, majorly spoken in South Asian countries, are nominative-accusative languages with person and honorificity agreement with the subject in all tenses, aspects, and moods. Typologically, EIALs exhibit free word order, with the canonical word order being SOV. There is no grammatical gender in these languages. EIALs are classifier languages without the presence of articles. The languages have both head-initial and head-final negation. Assamese, Bangla, and Odia are three major languages of the EIAL family with well-recorded diachronic text: *Caryapada*. *Caryapada* possessed the ergative marker *ẽ̄* seen in only past tense constructions. The current loss of ergative marker and the presence of split ergativity are common phenomena in EIALs. We use Assamese, Bangla and Odia as the representative Eastern Magadhan languages to show that all the EIALs have gone through the loss of ergativity despite their current synchronic idiosyncrasies. The subsequent subsection elaborates on the grammatical properties of *Caryapada* before moving on to the loss of ergativity seen in the present-day EIALs.

3.2 Grammar of *Caryapada*

Caryapada is the oldest text of the languages. It is a collection of poems written in the eighth – twelfth century. It is the only surviving text for early EIALs and was discovered by Haraprasad Sastri at the Royal Palace of Nepal in 1907. The source of the grammatical patterns of *Caryapada* is erstwhile MIA, in contrast to Old Indo-Aryan (OIA) (1500–1200 BC). OIA is rich in case markers and has a person-number-rich agreement system in all tenses and aspects (1) is in the present tense. For example, in (1), we see DPs like 'this you', a combination of demonstrative and second person plural *sá tvám*, which agrees with the verb 'smite'. (2) is a perfective construction, where we again see first-person singular nominative *ahám*. Verbal agreement displays the person-number phi-features of the nominative marked subject.

(1) Old Indo-Aryan
sá tvám agne vibhā́vasuḥ sṛján sū́ryo ná
this.NOM you.NOM Agni-VOC rich.in.light.NOM spreading.NOM sun.NOM like

raśmíbhiḥ śárdhan támāṃsi jighnase
beams.of.light.INS bold darkness.ACC smite.2SG.PRS
'(*The) you, Agni, rich in light like the sun spreading out with rays, boldly strike the darkness (away).' (Dahl, 2010, p. 146)

(2) *ahám etā́ mánave viśváścandrāḥ sugā apáś*
I.NOM these.ACC Manu.DAT all.glittering.ACC easy.to.traverse.ACC waters.ACC

cakara vájrabāhuḥ
make.1.SG.PRF mace.handed.NOM
'With the mace in my hand I made these all-glittering waters easy to traverse for Manu.' (Dahl, 2010, p. 143)

However, in MIA, the agreement system does not remain stationary when the case markers go through a quantitative change. Instead, we see manyfold changes in that arena. Unlike OIA, one of the most important agreement-related changes is that MIA does not participate in person-number agreements across all tenses and aspects. We see a rise of *-ta/-na* marked new perfective. The new perfective marker is **-ta/ *-na* is derived from the adjective of proto Indo-European and OIA. In this period, the newly formed perfective aspect becomes a vital factor. As from the late MIA, when it disintegrates into EIALs and its western counterpart Western IAL, we see more changes in the case system. One of the changes is an ergative system with separate nominative and ergative markers. The language also shows object agreement in its past/ perfective tense-aspect, as seen in (3). In (3), *hau* agrees with the verb in the present in person feature. However, *pai* in the same sentence does not agree with the verb. The first person *hau* is assigned the nominative case in the present with person agreement. However, the second person, *pai* gets an oblique case in non-present and does not trigger agreement. Later, this pattern is regularised with the rise of ergativity in non-present tenses. (see Deo, 2006 for a detailed discussion)

(3) Early MIA
hau pai pucchimi … diTThî pia pai sâmuha
I.NOM you.OBL ask.PRS.1SG seen.F.SG loved.F.SG you.OBL in-front

jant-i
passing.NOM-F.SG
'I ask you… Did you see (my) beloved passing in front (of you)?
(Vikramorvasiya; cited in Montaut, 2006)

A similar trend is seen in *Caryapada*. In the past and future, the agreement is with the object. In the present, however, the agreement is canonical. See Examples (4) and (5) below where the agreement system of (4) aligns with the feature specifications feminine gender of the object *bangali* rather than the masculine gender of the subject *bhushuk*. This sentence is in the past tense. However, in the case of (5), which is a sentence in the present tense, we see subject agreement between *birnad* and *bajae*.

(4) OEIA
aji bhushuk bangali bha-il-i
Today bhushuk.NOM inferior.being.F.SG become-PST-F.SG
'Today Bhushuk became bangali/ inferior.'

(5) *omaha damru baja-e birnad-e*
Uncalled damru beat-PRS.3 birnad-NOM
'Birnad beats the uncalled damru.' (*Caryapada*)

Now, we look at the case markers. *Caryapada* used different case markings for subjects in the present and past tense. (4) and (5) do not have the same case markers on the subjects. Before discussing the phenomenon in detail, we first list the major case markers of *Caryapada* in Table (4) below:

Table 4. Case markers of *Caryapada*

Case	**Markers**
Nominative	*-e*, *-o*, null
Accusative	*-e*, *-re*, null
Instrumental/ Ergative	**-*ẽ̃*, null**
Dative	*-ke/-ka, -ku -e*

In Table (4), the instrumental and ergative case (in bold) interest us for two reasons. First, we claim that though the overt markers are similar, the instrumental and ergative cases in *Caryapada* are licensed by different heads in the structure. Additionally, the ergative case -*ẽ̃* is related to the Voice head in this stage.

Before going into the analysis, we now look into some examples where this marker is seen. Chatterji (1926) identifies -ẽ̃ as an instrumental marker. Examples (6–7) attest to the fact that this marker is used for all adjunct PPs just as is expected of instrumental (adjunct) PPs. In (6) and (7), the marker -ẽ̃ comes in instrumental contexts with 'loud voice' and 'lion', respectively.

(6) OEIA
kalela sad-ẽ̃
loud voice-INS
'In a loud voice.' (*Caryapada;* cited in Sen, 2012)

(7) *siala sih-ẽ̃ sama jujha*
jackal lion-INS with fight
'The jackal fights with the lion.' (*Caryapada;* cited in Sen, 2012)

However, as mentioned above, these are not the only contexts where it can occur. For illustration, see (8). In (8), the -ẽ̃ marker can come with subjects of past tense, such as *kanh.*[2]

2. The source of the marker ẽ̃ and its connection with the past tense is seen prior to OEIA stage too. The same pattern is seen in the sentence (i) from the scripture of *Jaugada* (around 250 AD) in the Magadhan region in Eastern Prakrit in the Late MIA. They continue the tradition of MIA with the loss of rich nominative case endings with the agreement patterns. As seen in (i), which is a perfective construction, we see number and gender agreement with the feminine nominative object *dhammalipi*. Here, the person agreement is blocked in both instrumental case marked subjects such as *lajina* 'king' and the nominative object *dhammalipi* 'law scripture'. In contrast, number and gender agreement is seen between the nominative object *dhammalipi* and *lekhita* 'inscribed'. The morphological marker *-ena* is to be noted here. Such instrumental subjects are only noted in non-present constructions. Also see (3) for the contrast between nomiative subject and instrumental subject. *-ena* is the source of the OEIA marker ẽ̃ (see Chatterji, 1926; Montaut, 2006 for discussions).

(i) *iyam dhammalipi divanampiy-ena piyadassina*
this.NOM.F.SG law.scripture.NOM.F.SG god.friend-INS.M.SG friendly.looking.INS.M.SG
(lajina) lekhita
king.INS.M.SG inscribed.NOM.F.SG
'The friendly looking king beloved of gods has (made) engraved the law-edict.'
Eastern Prakrit (Montaut, 2006)

Deo (2006) shows the tense-based system of OIA changed to the aspect-based system during MIA. It means that participial *-ta/-na* of OIA is primarily interpreted as the new perfective and the present as imperfective. One of the changes is the rise of the ergative system, with separate nominative and ergative markers cropping up.

(8) OEIA
kanh-ẽ̃ pothi padh-il-i
kanha-ERG book.F read-PST-F
'Kanha read a book.' (*Caryapada;* cited in Chatterji, 1926, p. 742)

According to Sen (2012), this *ẽ̃* marker is passive. However, Deo (2006) notes that the passive to active transformation was already completed in the preceding MIA due to the innovation of a new perfective, as discussed above.

Additionally, in ergative constructions, as seen in (8), we see object agreement with the feminine gender of *pothi*. In contrast, neither the *-ẽ̃* marker on the subject nor the object agreement is seen with constructions in the present tense, as seen in (9). Here the subject of the unaccusative verb 'say' takes the nominative marker *-e*.

(9) *bhad-e bhan-ai*
bhada-NOM say-PRS.3
'Bhada says.'

If we compare (8) with its minimal pair in (10), we notice the difference in the case markers more clearly. The case marker on the subject DP is *ẽ̃* in (8), whereas it is *e* in (10). Example (10) is in the present tense. There is also variation in agreement on the verb 'read' as we see object agreement in (8) and subject agreement in (10).

(10) *kanh-e pothi padh-ai*
kanhe-NOM book.F read-PRS.3
'Kanha reads a book.' (Chatterji, 1926, p. 742)

However, the languages lost their ergative patterns with time. The following section elaborates on it.

3.3 Loss of *-ẽ̃*

The loss of *-ẽ̃* and object agreement is seen later in LMB and as early as Old Odia and Early Assamese. In LMB, the *-ẽ̃* marker and object agreement were lost, as deduced from the examples below.

(11) LMB
...dake shob lok-e
call.PST.HAB-3 all people-NOM
'All people called.' (*Baishnabpadabali)*

In (11), we see the sentence is in the past tense. However, there is subject agreement and a nominative marker. No examples of *-ẽ̃* and object agreement are found in *Baishnabpadabali*, which is an LMB text. Similar examples from Old Odia

(12–16) and Early Assamese (17) are seen below. Old Odia data (12–13) are from the *Bhuvaneswara Stone Inscriptions* (in *Samvat 22* of *Vira Narasimhadeva*), as seen in Tripathi (1963). Examples in (14–15) from the Middle Odia text *Srimadbhagabat*, as presented in Tripathi (1963), illustrate similar patterns. Old Assamese (see 17) is from the Early Assamese text *Prahlada Caritra*, according to Devi (1986).

(12) Old Odia
acay-e pani dhi-l-a
acharya-NOM water give-PST-3.SG
'Acarya gave water.' (*Bhuvaneswara Stone Inscriptions*)

(13) Old Odia
svahast-ẽ̃
Own.hand-INS
'By own hand.' (*Bhuvaneswara Stone Inscriptions*)

(14) Fifteenth century Odia
*Muni-***e** *ekant-e bichaari*
Saint-PL loneliness-in consider.3.PRS
'Saints consider/discuss alone.' (*Srimadbhagabat* ch 1, p. 31)

(15) Fifteenth century Odia
sru-jan-e piba anukshyane
great-man-PL drink.IMP carefully
'Great men drink it carefully.' (*Srimadbhagabat* ch 1, p. 31)

(16) Fifteenth century Odia
*aahe budha jan-***e** *mora na dharibaa dosa*
oh wise man-PL I.GEN not hold fault.
'Oh! wise people, don't hold/remember my fault.' (*Candipurana:4*)

(17) Early Assamese
mor-e mati-ss-e
crow-NOM call-PROG-3
'The crow is crowing.' (*Prahlada Caritra;* cited in Devi, 1986)

Additionally, the marker and the object agreement are also not seen in present-day languages (as shown in Examples (18–21)).

(18) Present-day Odia[3]
jon mili-ku dekh-il-aa
john.NOM Mili-ACC see-PST-3SG
'John saw Mili.'

(19) *se nach-il-aa*
he.NOM dance.PST-3SG
'He danced.'

(20) *se pad-il-aa*
he.NOM fall-PST-3SG
'He fell.'

(21) *se kalama-re/dvaraa lekhu-ch-i*
he.NOM pen-INS write-PRS.PROG-3SG
'He is writing with a pen.'

Since we have seen the changes in the EIALS in this section, the subsequent section analyzes this diachronic trajectory. We claim that the change from the ergative system to the nominative-accusative system occurs because of the loss of the Voice head in the derivation.

4. Voice and -*ẽ̃* marker

4.1 The Voice head

This section shows evidence of the presence of the Voice head inside the Voice Phrase in structures in the past tense in *Caryapada*. The primary evidence is the lack of subject agreement in past-tense constructions in *Caryapada*. This strongly implies that T or past tense is not participating in agreement with the ergative marked subject (see Examples (8) and (10)) and is not responsible for the ergativ-

3. Compare with Hindi-Urdu, a Western IAL, which is a split-ergative language synchronically. It exhibits ergative marker *ne* with object agreement in perfective aspect in (ii), in contrast to (i).

(i) *raam-ne roti khaayi*
ram-ERG bread.F eat.PVF.F
'Ram ate bread.'

(ii) *raam roti khata hai*
ram.NOM bread.F eat.M be.PRS.HAB.3.SG
'Ram eats bread.'

ity seen in *Caryapada*. This pushes us to understand Voice, as a functional head, plays an important role in the structure. The Voice head is the locus of the ergative case (see Legate, 2014, among others). Chomsky (1995) introduces a functional head *v* over VP, which introduces the external agent. This is an attempt to describe the verbal domain as a layered one where the VP is layered by an extra functional head. Kratzer (1996) additionally introduces VoiceP over *v*P. She claims that all the external arguments are introduced higher in the [Spec, VoiceP] instead of *v*P. Kratzer's Voice is intricately related to argument structure and theta role. In connection to the introduction of Voice head, Kratzer notes the difference between eventive (agent) external arguments and stative (experiential) external arguments. She proposes that the Voice head's characteristic is contingent on the verb type and the theta role it assigns. 'Mittie' of (22) is the agentive external argument and is generated in the [Spec, VoiceP], unlike the non-agentive arguments. This is illustrated in Figure 1.

(22) Mittie fed the dog.

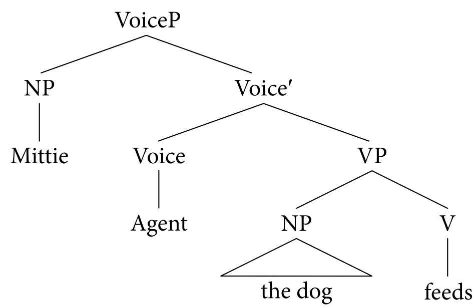

Figure 1. Structural representation of (22)

In the Minimalist literature, the intricate layer of the verbal domain has been established via the presence of ApplP (see McGinnis, 1998, 2001), Inner Aspect (see Macdonald, 2006), Passive P and Middle V (see Sundaresan and McFadden, 2017), CausP (see Key, 2013), among others. Additionally, the distinction between *v* and Voice has been argued for by many scholars like, Sundaresan and McFadden (2017) in Tamil, among others.

The importance of the extra layer over *v*P can also be seen in connection to ergativity. Ergative literature, such as Aldridge (2004, 2008), differentiates *v* in Tagalog, which can assign ergative case from the one which cannot. First, transitive *v* has the EPP feature in syntactic ergative languages. This makes the absolutive object move out of *v*P in the edge for further operations. The movement out of the *v*P helps the argument to be visible to the probe in a higher phase, for example, as seen in Figure 2 for sentence (23).

(23) Tagalog
ano ang b-in-ili ng babae?
what ABS TR-PRF-buy ERG woman
'What did Maria buy?' (Aldridge, 2008)

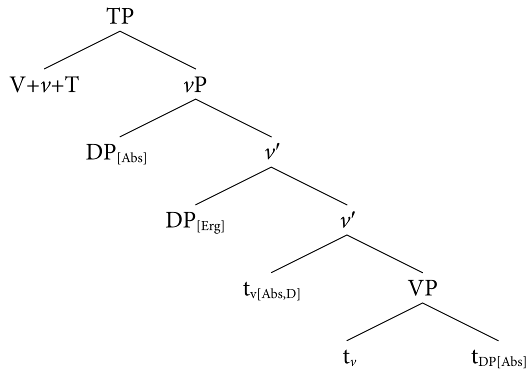

Figure 2. Structural representation of (23) (Aldridge, 2008)

Along with its relation to the external argument, the ergative argument behaves differently from the non-ergative one vis-a-vis the structural position and its relation to the functional head. In Figure 2, the extra layer is seen via the presence of extra specs. The intransitive *v* has no EPP feature. Therefore, the object DP can move out of *v*P.

Legate (2002) also claims that transitive *v* is responsible for assigning ergative in ergative languages and accusative in accusative-absolutive languages. This transitive *v* or Voice head assigns an inherent ergative case to its specifier. It does not depend on the structural position of the DP. Legate (2017, p. 182) mentions that "this head is sometimes also referred to as Voice". Works like Dixon (1994), Authier and Haude (2012), Coon (2017), among others, also highlight and establish the relationship between Voice and ergativity. The next section highlights Polinsky (2016) to show how Voice plays a crucial role in the loss of ergativity in EIALs.

4.2 Voice head and ergativity (Polinsky, 2016)

Polinsky (2016) distinguishes the DP-ergative and the PP-ergative in ergative languages. Before we use this distinction to examine the ergative constructions in Caryapada, we briefly explain the concept of DP-ergative and the PP-ergative and the related diagnostics below.

The rudimentary distinction between DP-ergative and PP-ergative is that the ergative case of the former is a structural case, and that of the latter is an inherent case. The functional head Voice that licenses the external argument (EA), assigns the ergative case in DP-ergatives. On the other hand, the ergative case in PP-ergatives is assigned by an (overt or silent) adposition P. In the case of a DP-ergative (shown in Figure 3), the lexical VP consisting of the object DP merges with the lower functional head *v*. This *v* assigns an absolutive case to the object DP. VoiceP is merged with the lower functional head, *v*. The external argument is merged in the specifier position of VoiceP. The *v* head assigns theta role, and the Voice head assigns the structural ergative case to the external argument.

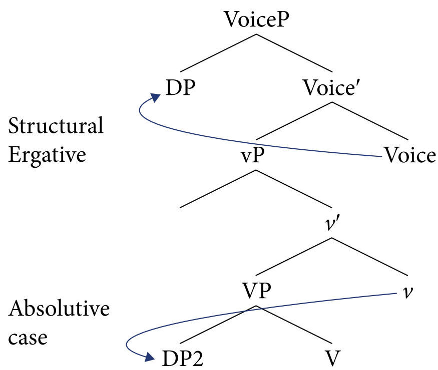

Figure 3. Structural representation of DP-ergativity

PP-ergative (shown in Figure 4) is assigned by the P head. The lexical V and object DP are merged to form VP. VP merges with the lower functional head *v*, which assigns the absolutive case to the object DP. Voice head, the higher functional head in the verbal domain is merged next in the structure. PP is merged in the specifier of VoiceP. PP is constituted of P and the external argument DP. P assigns the inherent ergative case to the external argument.

We follow the diagnostics used by Polinsky (2016) to base our argument. The list of diagnostics are agreement, A-movement, A′-movement, Binding and Co-ordinate structures. We elaborate on the diagnostics below, as described in Polinsky (2016).

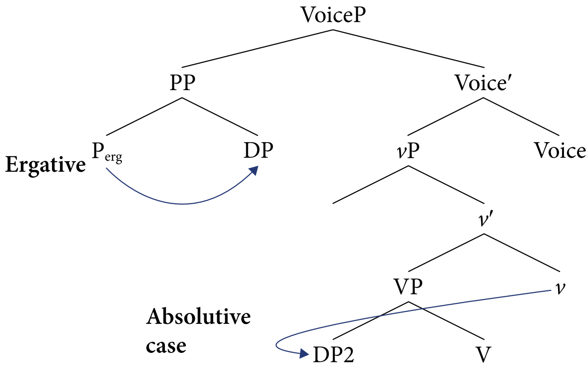

Figure 4. Structural representation of PP-ergativity

i. *Agreement*

DP-ergative triggers agreement, in contrast to PP-ergative. (24) and (25) are examples where DP-ergative in Chol participates in agreement, whereas PP-ergative does not participate in agreement, as seen in (26) in Tongan. The natural after-effect, according to Polinsky, is that DP-ergative is structural, PP-ergative is not.[4]

(24) Tabasaran
pro aldaq^{h}n-uv
ABS fall-3SBJ
'He/She/it fell.'

(25) *pro pro uvc^{w}un-uv*
ABS ERG beat-3SBJ
'He/She beat him/ her.' (Polinsky, 2016, p. 132)

(26) Tongan
nae 'ikai (ke) 'ilo 'e Siale 'a ha me'a
PST NEG SBJV see ERG Charlie ABS DET thing
'Charlie did not find a thing.' (Lit: It was not that Charlie found a thing)
(Polinsky, 2016, p. 189)

Now, we move on to the second diagnostic, which is A-movement.

4. There are some exceptions found in case of PP agreement. See Polinsky (2016) for a review and discussion.

ii. *A-movement*

The DP-ergative subject can move to case-related positions in the structure, i.e., argument positions. In other words, it can participate in A-movement. However, the PP-ergative fails to participate in A-movement, such as raising. This is illustrated using Tsez, a DP-ergative language (as shown in (27) and (28)), and Tlingit, a PP-ergative language (as shown in (29) and (30)). (28) is the raised version of (27), where the ergative subject 'the bird' is raised out of the original clause. In (27), the infinitival clause and the matrix verb agree in gender class IV. Contrastingly, in (28), the matrix verb agrees with the raised subject the 'bird' in gender III.

(27) Tsez
[nela ay-a ko y-ac-a] r-ici-xosi yol
DEM bird-ERG raspberry.ABS.II II-eat.TR-INF.IV IV-stay-PRS.PTSP be.PRS
'This bird keeps eating the raspberries.' (Polinsky, 2016, p. 317)

(28) *za ayi [ko y-ac-a] b-ici-xosi yol*
DEM bird.ABS.III raspberry.ABS.II II-eat.TR-INF III-stay-PRS.PTSP be.PRS
'This bird keeps eating the raspberries.' (Polinsky, 2016, p. 317)

Tlingit, a PP-ergative language, on the other hand, does not allow raising, a type of A-movement. It has no preposition to express raising, as seen in other languages, such as English (in the case of the verb 'seem'). It does not have modal or aspectual verbs and concepts related to aspect are expressed via pre-verbal particles. (29) and (30) show the particles related to 'begin' and 'finish' *gunei* and *yan*, respectively.

(29) Tlingit
gunei uwakux
INCEPT PF.3SG.SBJ.go.by.vehicle
'He started driving.'

(30) *yan at xwaaxaa*
TERMINATIVE PFV 1SG.SBJ.eat
'I've finished eating.' (Polinsky, 2016, p. 103)

Therefore, according to Polinsky, Tabasaran, a DP-ergative language can participate in A-movement, in contrast to Tlingit, a PP-ergative language. The next diagnostic is A′ movement.

iii. *A′ movement*

DP-ergative language allows A′ movement in contrast to PP-ergative language. For example, in Tsez A′ movement is allowed making it a DP-ergative language. DP-ergatives participate in the formation of the relative clauses. In (32), the ergative subject 'boy' of (31) is relativized. Tongan, as seen in (33), does not allow ergative extraction with a gap and leaves a resumptive pronoun *ne* at the extraction site. It does not allow A′ movement.

(31) Tsez
Uz-a kayat kid-be-r teƛ-si/teƛ-xo
boy-ERG letter.ABS.II girl-OS-LAT give-PST.WIT/give-PRS
'The boy gave/gives a letter to the girl.'

(32) [_$_i$*kayat kid-be-r-no taʎ-ru] uzi*$_i$
letter.ABS.II girl-OS-LAT-TOP give-PST.PTCP boy.ABS.I
'The boy that, to the girl, gave a letter.' (Tsez. Polinsky, 2016, p. 313)

(33) Tongan
*'a e ta'ahine ['oku *(ne) tanaki nai 'a e sitapa*
ABS DET girl PRS RP collect PRT ABS DET stamp
'The girl that collects stamps.' (Polinsky, 2016, p. 149)

Now. we show the diagnostic of binding and how it works differently in DP-ergative and PP-ergative languages.

iv. *Binding*

Reciprocals and reflexives can be A-bound in DP-ergatives, but PP-ergative subjects can not A-bind. To illustrate, Polinsky (2016) shows that Tsaxur has a dedicated anaphor with ergative subjects (which is incompatible with absolutive subjects), as seen in (34). She further, referring to Mosel and Hovdhaugen (1992), states that Samoan has no reflexive constructions. They express reflexivity using lexically specified reflexives like *ta'ele* 'bathe', labile causative verbs such as *fa'apa'u* 'fell', etc. It indicates Samoan's inability to participate in A-binding.

(34) Tsaxur
rasul-e wuʒa get-u
rasul-ERG self-ABS CLASS1.beat-PRF
'Rasul beat herself up' (Polinsky, 2016, p. 135)

(35) Samoan
na fa'a-pa'u le teine i le moega
PST CAUS-fall DET girl LOC DET bed
'The girl threw herself onto the bed'
(Polinsky, 2016, p. 95; cited in Mosel and Hovdhaugen, 1992)

The last diagnostic we use from Polisnky is Co-ordinate structures.

v. *Co-ordinate structures*

In DP-ergative subject constructions, the ergative case appears on both conjuncts. By contrast, the PP-ergative marker appears only once on the entire coordinate phrase. As cited in Polinsky, the Hindi ergative marker *-ne* joins the entire coordinate phrase, as seen in (36), making Hindi ergative a PP-type ergative (cf. Mahajan 1997, Mohanan 1995). Patterns such as (37) will be grammatical for a DP-ergative language.

(36) Hindi-Urdu
[larki aur larkaa]-ne
boy and girl-ERG

(37) **larki-ne aur larka-ne*
boy-ERG and girl-ERG
'Boy and girl'[5]

Now, we apply these diagnostics to ergative constructions in *Caryapada*, which we have discussed in Section 3, to ascertain the status of ergativity in the language in the next sub-section.

4.3 Voice and ergativity in EIALs

This section discusses how Voice head is related to the presence of ergativity in *Caryapada*. When we apply the first diagnostic of agreement, the ergative-subject triggered agreement is blocked in *Caryapada*, as seen in the earlier sections. Example (10) is repeated below as (38).

(38) *kanh-e pothi padh-ai*
kanhe-NOM book.F read-PRS.3
'Kanha reads a book.' (Chatterji, 1926, p. 742)

However, in the past tense, we see the case marker and the alignment changes, as seen in Section 2.

(39) OEIA
kanh-ẽ̃ pothi padh-il-i
kanhe-ERG book.F read-PST-F
'Kanha read a book.' (*Caryapada*, Chatterji, 1926, p. 742)

This diagnostic shows that *Caryapada* ergativity behaves similarly to a PP-ergative one.

5. Some exceptions are also discussed in Polinsky (2016).

We do not find evidence for A-movement in *Caryapada*.

For the diagnostic, A′ movement, ergative subjects can participate in Wh-questions formation, as seen in (40).

(40) OEIA
tabe to-ka rakhi-b-a kona jan-ẽ̃
then you-ACC protect-FUT-GER.F which person-ERG
'Then who will protect you?'
(*Caryapada;* cited in Montaut 2016; glosses slightly modified)

Now, we move to the next diagnostic, which is binding. In this diagnostic, the binding between anaphor *apna* and the third-person covert DP is seen.

(41) OEIA
puno loiya apna chatar-iyu
(he) again taking himself/herself kill-PST.3
'taking zero/ again, he killed himself.' (*Caryapada* 26)

Another piece of evidence comes from co-ordinate structures, as seen in (42) where *the marker ẽ̃* occurs on both the DP in a co-ordinate structure, unlike present Western IALs such as Hindi-Urdu.

(42) OEIA
ali-ẽ̃ kali-ẽ̃ bat rundh-el-a
ali-ERG kali-ERG path block-PST-M
'Ali and Kali blocked the path.' (*Caryapada* 7)

We summarise our findings in the table below.

Table 5. Summary of diagnostics which show *Caryapada* showed DP-ergative patterns

Diagnostics →	**Agreement**	**A- movement**	**A′ movement**	**Binding**	**Co-ordinate**
DP-Ergative	✓	✓	✓	✓	✓
PP-Ergative	✗	✗	✗	✗	✗
Caryapada	✗	–	✓	✓	✓

Following Polinsky's (2016) analysis, as explained above and summarised in Table (5), evidence indicates that *Caryapada* had DP-ergative properties. This implies the presence of a functional head Voice, which licenses the external argument with the ergative case. For a sentence, such as (39), the representation can be seen in Figure 5. Here, we see that the DP *kanhẽ̃* gets the structural ergative case from an extra layer over *v*P, the VoiceP.

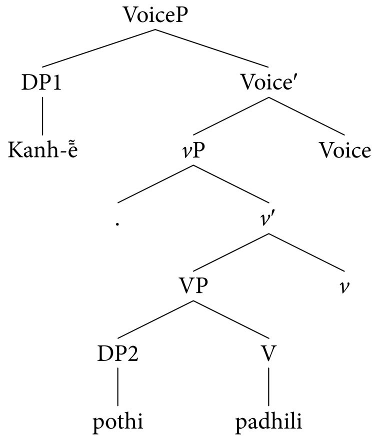

Figure 5. Structural Representation of DP-ergativity in *Caryapada*

Therefore, this section establishes that the Voice head is responsible for the ergative patterns we see in *Caryapada*.[6]

Section 3.2 above shows that the ergative marker and the related object agreement were lost in LMB. We can connect the loss of ergativity to the loss of Voice head in the languages. This is the reason why the EIALs convert to a nominative-accusative language family from a split ergative one. This change is not seen in its sister language, the Western IALs (the representative language being Hindi-Urdu).[7]

More evidence of the loss of VoiceP in the later stages comes from the loss of canonical passives in such languages. *Caryapada* exhibited productive passives, as seen in (43) and (44).

6. This pattern was seen till Early Middle Bangla (see Table 3).

7. In the present tense, the subject agreement is an important diagnostic where the external argument is related to T/ Infl in all tenses. For illustration, see examples (i) and (ii) below where both the first and the second person trigger agreement.

(i) *ahme deh-u*
we.NOM give-PRS.1
'We give.' (*Caryapada* 19; cited in Chatterji, 1926)

(ii) *baha-tu*
Row-2
'Row thou.' (*Caryapada* 8; cited in Chatterji, 1926)

(43) OEIA
harina harinira nila na jan-i(a)
stag doe-GEN abode NEG know-PST.PTCP
'the abode of the stag and doe is not known.' (*Caryapada*)

(44) *areshori jai*
reach PST.PTCP
'(bank) is reached by one.' (*Caryapada*)

Examples (43) and (44) show evidence of synthetic and analytic passive from *Caryapada*, respectively. Both were gradually lost in the language. We claim that this change happens because of the change of Voice head. Transitive *v* lost the Voice head and its specifier in later periods and synchronic variants. Therefore, no canonical passives are seen.

In Modern EIA languages, the clausal structure looks like Figure (6).

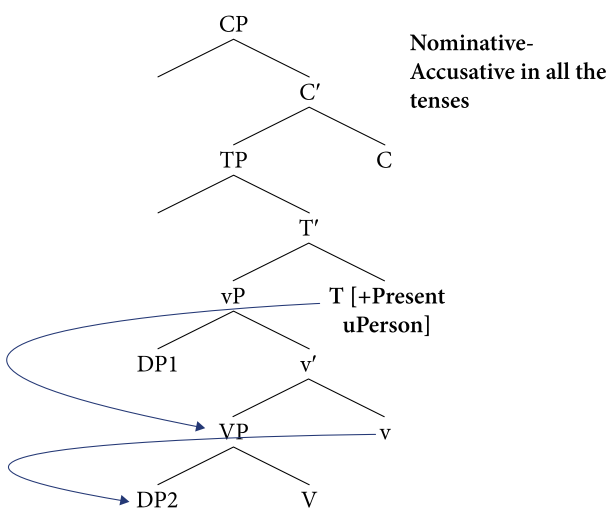

Figure 6. Structural representation of new EIA languages

The head T licenses external arguments in transitives and internal arguments of unaccusatives uniformly. The Voice head, which is seen in Figure (5), is absent in Figure (6) over the *v*P layer.

5. Conclusion

In this paper, we have shown how the ergative *ẽ* marker and agreement system has undergone a change in EIALs diachronically. This ergative marker, which was present in *Caryapada*, was lost in the later stages, viz. Late Middle Bangla, Old Odia, and Early Assamese. Along with the loss of *ẽ* marker, we highlight the loss of ergativity through the loss of object agreement in these stages. According to our analysis, the change is due to the loss of the functional Voice head. This particular head was responsible for assigning DP ergativity to subject DPs in *Caryapada*. With the loss of this Voice head, the split ergative case and agreement pattern was replaced by the nominative-accusative one. In the present EIALs, the loss of passives also holds up the loss of Voice head. Therefore, this paper presents a detailed description of how ergativity was lost and replaced in Eastern Indo-Aryan languages.

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